

MATTHEW COX

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Education

Massachusetts Institute of Technology

PhD in Chemical Engineering

Expected May 2029

Cambridge, MA

California Institute of Technology

BS in Chemical Engineering (Computational Track); GPA: 4.1/4.0

June 2024

Pasadena, CA

Professional Experience

D.E. Shaw Research

June 2023 – September 2023

Chemistry Intern

New York, NY

- Developed predictive models using PyTorch-based graph neural networks for the accurate determination of atomic properties in drug discovery applications
- Applied meta- and transfer learning strategies to enhance model accuracy across molecular datasets, overcoming challenges associated with data scarcity
- Authored and executed parallelized workflow scripts in Bash for high-throughput density functional theory computations, significantly expanding the molecular dataset for model training

Protomer Technologies/Eli Lilly and Company

June 2022 – March 2023

Research Intern

Pasadena, CA

- Designed a biochemical assay for quantifying small molecule interactions with molecular sensors and generated a comprehensive experimental dataset using this assay
- Programmed a specialized Python package for the analysis of experimental data, streamlining the determination of binding affinities to support rapid sensor optimization
- Optimized assay protocols, significantly improving its throughput and accuracy

Research Experience

Massachusetts Institute of Technology

September 2024 – Present

Graduate Researcher, Coley Research Group

Cambridge, MA

- Ongoing research on post-training LLMs and building LLM agents for computational chemistry and materials science

California Institute of Technology

June 2021 – August 2024

Undergraduate Researcher, Hadt Lab

Pasadena, CA

- Developed a novel computational framework for the prediction of electron spin-spin relaxation rates in molecules and solids, combining quantum dynamics calculations with machine learning techniques
- Utilized Caltech's high-performance computing cluster for cluster-correlation expansion theory simulations to develop new models for qubit decoherence

Undergraduate Researcher, Goddard Lab

Pasadena, CA

- Designed a new approach for the simulation of surface Pourbaix diagrams, enabling fast prediction of the surface states of materials in aqueous environments
- Integrated Monte Carlo sampling and density functional theory calculations with an active learning algorithm

Undergraduate Researcher, Wang Lab

Pasadena, CA

- Implemented a graph-based genetic algorithm in Julia for the rapid optimization of molecular structures for chemical engineering applications

John Stauffer Summer Undergraduate Research Fellow, Hadt Lab

Pasadena, CA

- Analyzed electron spin-lattice relaxation rates in nitroxyl radicals, identifying the impact of spin-phonon coupling
- Performed computational analyses using Python, Mathematica, and ORCA, coupled with experimental synthesis and molecular characterization using mass spectrometry, NMR, and EPR techniques

Teaching Experience

California Institute of Technology

January 2023 – June 2024

Teaching Assistant for ChE 105/CDS 110, ChE 62, & ChE 15

Pasadena, CA

- Facilitated student learning in Dynamics and Control of Chemical Systems by guiding the application of Matlab for complex chemical engineering analyses and system control tasks
- Helped lead instruction in Introduction to Chemical Engineering Computation, emphasizing the practical use of computational tools (primarily Python) in chemical engineering

Technical Skills

Languages: Python (PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, RDKit), Julia, Mathematica, Matlab, Bash

Scientific Software: ORCA, VASP, Psi4, Avogadro, pyMOL, LAMMPS, AmberTools, VMD

Tools: Git, LaTeX, VS Code